Question **6**

Answer saved

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Consider the system of linear equations

$x+y-z=2$

$3x+2y-z=3$

$-x-y+2z=-1$

Write down the system in matrix form,

$AX=b$ where $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Write down the coefficient matrix A of the above system. Determine the inverse A^{-1} of A by applying the matrix inverse algorithm, that is by applying the sequence of elementary row operations in the sense $[A|I_3]$

where I_3 is the 3 by 3 identity matrix.

- ( ) $A=\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & -1 \\ 0 & 1 & 0 & 5 & -1 & -2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$
- (•) None of the given options are correct.
- ( ) $A=\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -1 \\ 0 & 1 & 0 & 5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$
- ( ) $A=\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & -1 \\ 0 & 1 & 0 & -5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$
- ( ) $A=\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 1 \\ 0 & 1 & 0 & 5 & -1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$